SECOND SEMESTER — LAB 1B ASSIGNMENT

FULL NAME: BANJO AYODELE OREOLUWA

REGISTRATION NUMBER: C11/26/FCDF/17169

COURSE CODE: CIP-B101

DATE: 6 NOVEMBER 2026

PLATFORM USED: WINDOWS 11 PC

LAB 1B: PC SYSTEM FUNDAMENTALS, DISK SECTORS AND PARTITION ANALYSIS

INTRODUCTION

This lab examines basic computer information and evidence found in Windows.

I checked the system specifications, storage devices and partitions.

I observed running processes and their process IDs and thread counts.

I calculated disk and partition sizes using sector information.

I also examined the boot process and file systems used on the PC.

These tasks show how computer records can help in digital forensic investigations.

TOOLS USED

Windows 11 Pro; System Information (msinfo32); Command Prompt (systeminfo, tasklist and DiskPart listing commands); PowerShell (Get-Disk, Get-Partition, Get-Volume and Get-Process); Disk Management.

PART A — SYSTEM IDENTIFICATION AND EVIDENCE LAYER MAPPING

A1. Capture basic system specification

Field	Observed value	Evidence
Operating system	Microsoft Windows 11 Pro; version 10.0.26200 (build 26200)	Figures 1, 3
Computer manufacturer/model	HP EliteBook 820 G3	Figures 1, 3
System architecture	x64-based PC	Figures 1, 3

Processor	Intel Core i7-6600U @ 2.60 GHz; 2 cores, 4 logical processors (as displayed in System Information)	Figure 1
Installed physical RAM	16.0 GB (System Information); 16,265 MB total physical memory (systeminfo)	Figures 2, 3
Firmware boot mode	UEFI	Figure 1
Secure Boot state	On	Figure 1
BIOS version/date	HP N75 Ver. 01.57, 28 July 2022	Figures 1, 3

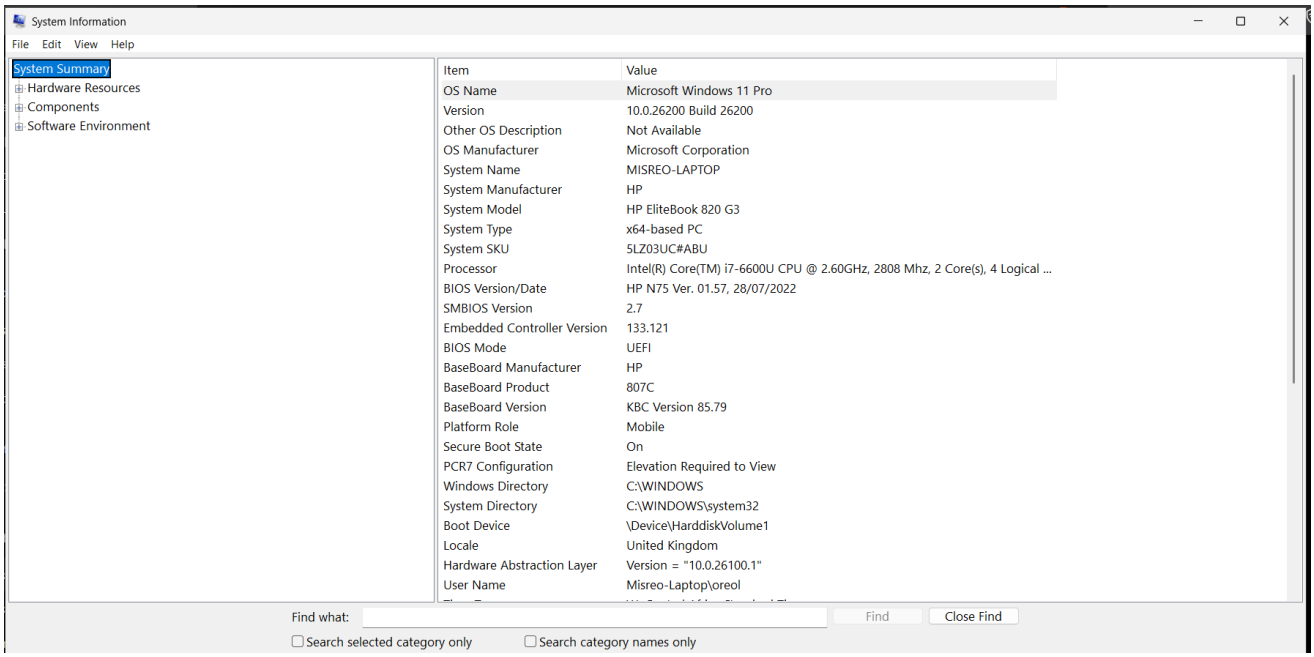


Figure 1. System Information: OS, computer model, processor, UEFI boot mode and Secure Boot state.

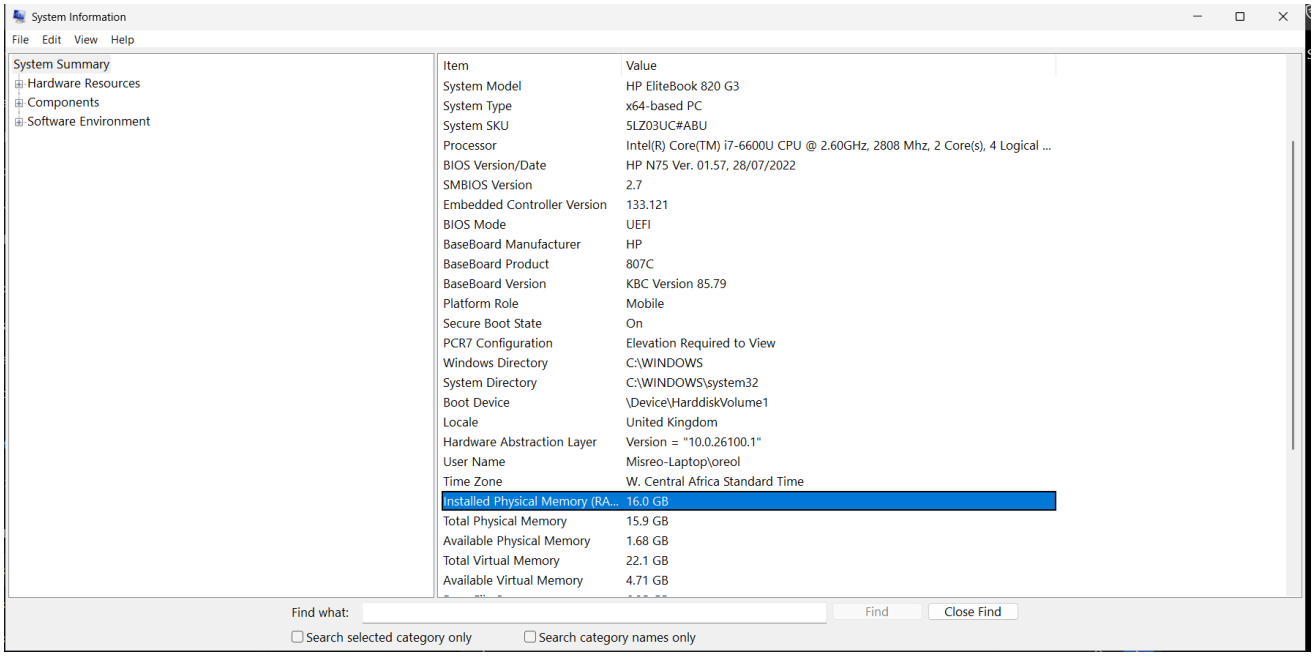


Figure 2. System Information: installed and available physical memory.

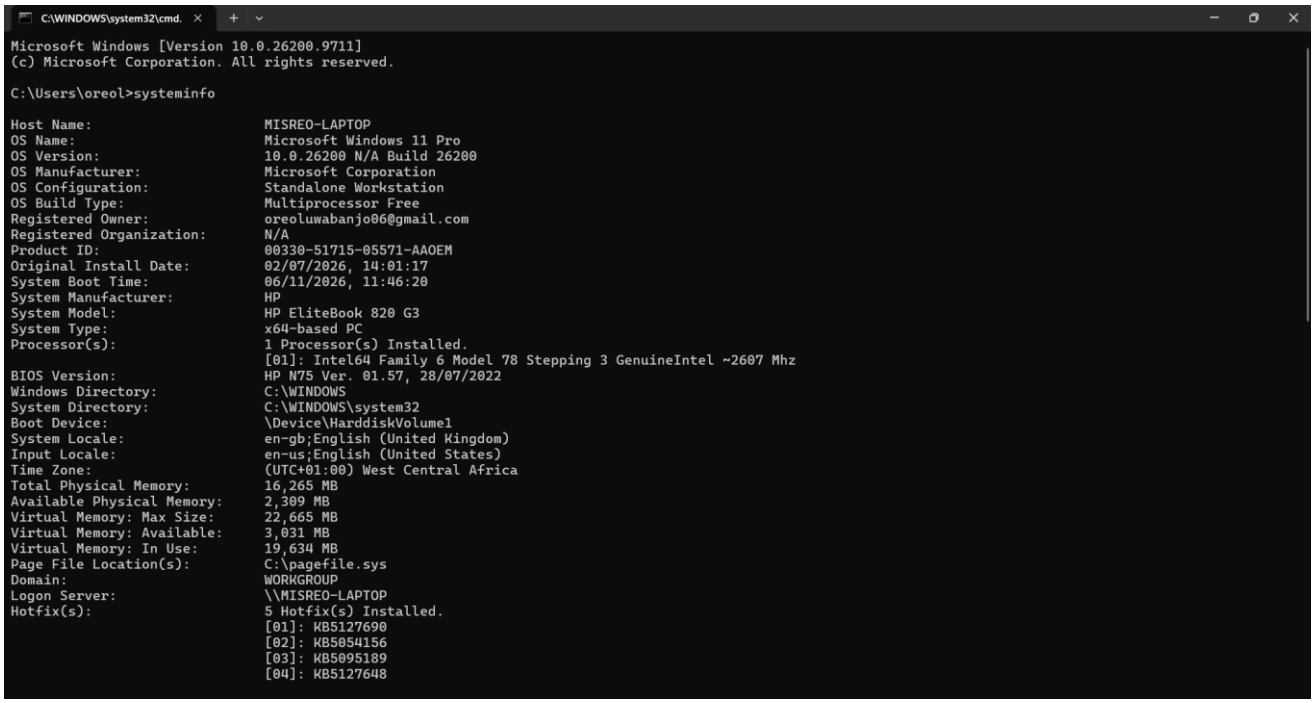


Figure 3. Command Prompt: systeminfo output confirming OS and computer specifications.

A2. Capture storage information

Disk property	Observed value
Disk number	0
Model	INTEL SSDSCKKF256G8H
Disk capacity	256,060,514,304 bytes (256.060514304 decimal GB; approximately 238.47 GiB)
Partition style	GPT
Logical sector size	512 bytes

Physical sector size	512 bytes

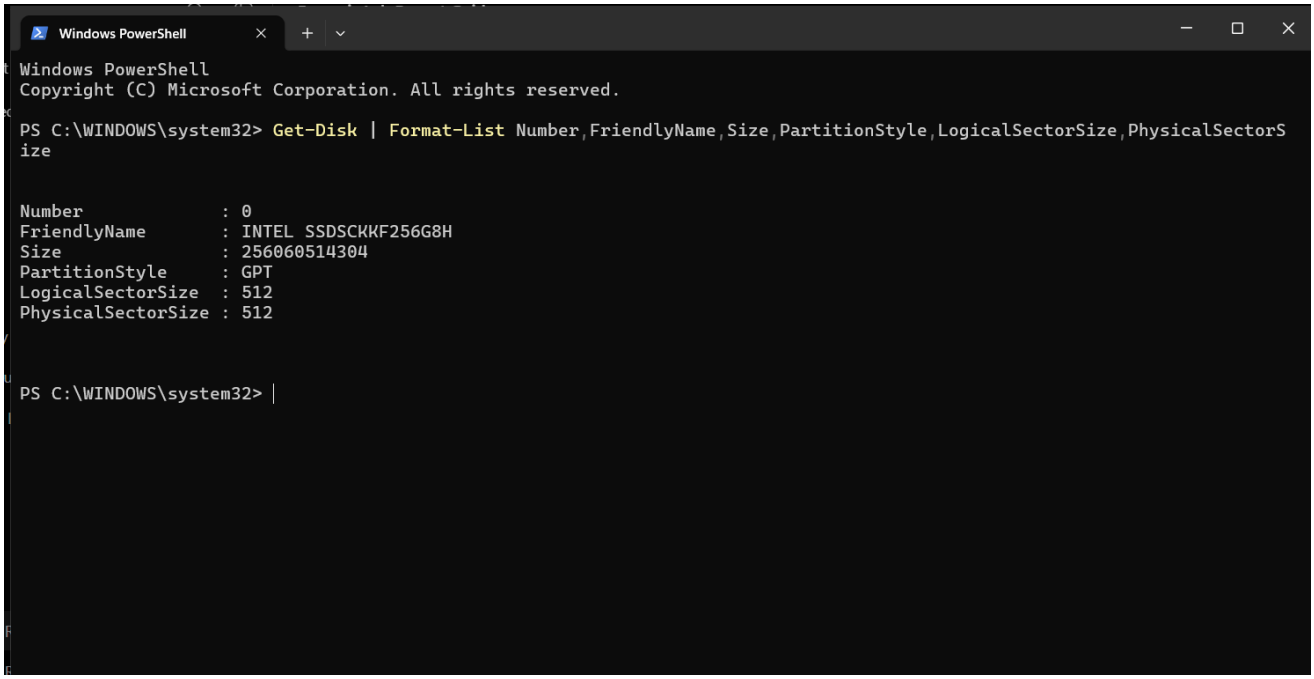


Figure 4. Get-Disk output showing model, capacity, GPT and sector sizes.

No.	Type	Offset (bytes)	Size (bytes)
1	System	1,048,576	209,715,200
2	Reserved	210,763,776	16,777,216
3	Basic (C:)	227,540,992	254,868,979,712
4	Recovery	255,096,520,704	962,592,768

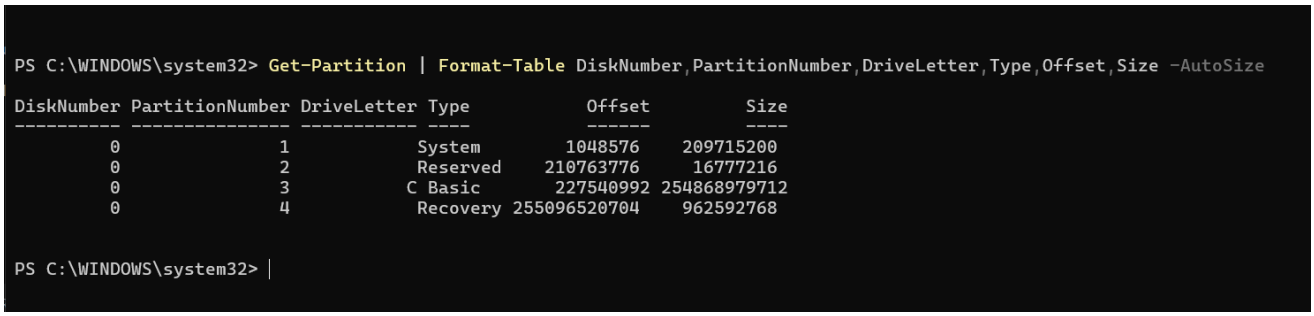


Figure 5. Get-Partition output listing Disk 0 partition numbers, types, offsets and sizes.

Volume	File system	Volume size (bytes)	Free space (bytes)
C:	NTFS	254,868,975,616	75,482,394,624
No drive letter	NTFS	962,588,672	120,283,136

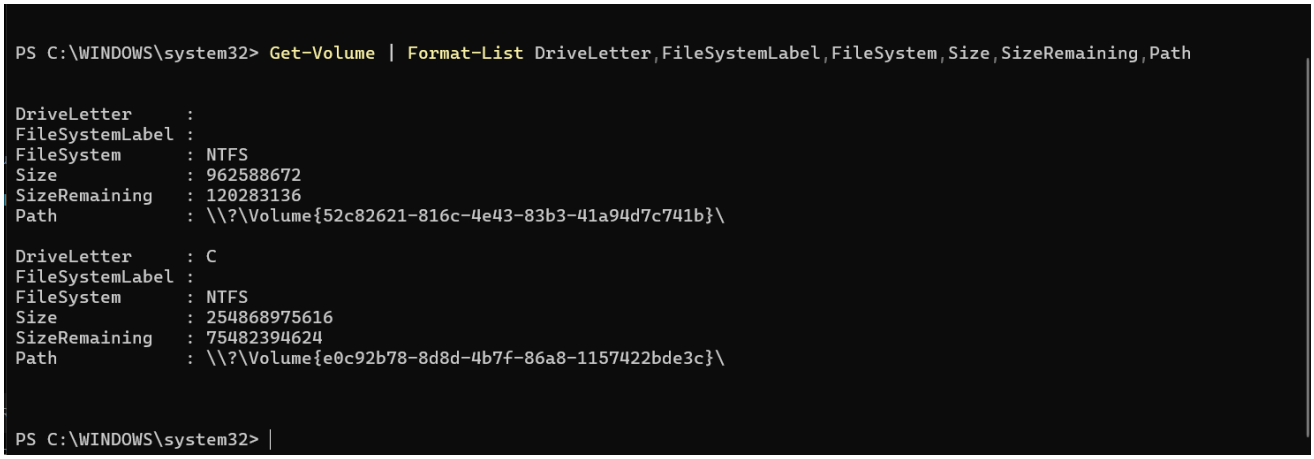


Figure 6. Get-Volume output showing NTFS file systems, volume sizes and remaining space.

A3. Map evidence sources to system layers

Layer / source	Example evidence	Where it may be found	Forensic importance
User layer	Opened files, search terms, account activity	Recent-file lists, browser history, user profile	Associates actions with a user context; attribution requires corroboration
Application layer	Document metadata, browser cache, application logs	Application databases, profile folders, logs	Identifies software use and potential activity traces
Operating system layer	Event records, installed software, user accounts	Windows Event Logs, Registry, system directories	Helps reconstruct system events and account activity
Disk / file-system layer	Files, deleted entries, metadata, partitions	NTFS metadata, allocated and unallocated disk space	Supports file recovery, provenance and timeline analysis
Memory / process layer	Active processes, PIDs, transient data	Live process listings and authorized RAM acquisition	Can reveal volatile execution state not preserved on disk

The HP EliteBook 820 G3 runs Windows 11 Pro with 16 GB RAM, UEFI and Secure Boot. Its Intel SSD uses GPT and contains four partitions. Two NTFS volumes were reported.

PART B — PROGRAM, PROCESS AND THREAD INVESTIGATION

B1. Observe running processes

Process name	PID	Likely purpose
smss.exe	560	Windows Session Manager; helps set up system sessions.
winlogon.exe	924	Windows logon process; supports sign-in and sign-out.
services.exe	996	Service Control Manager; manages Windows services.

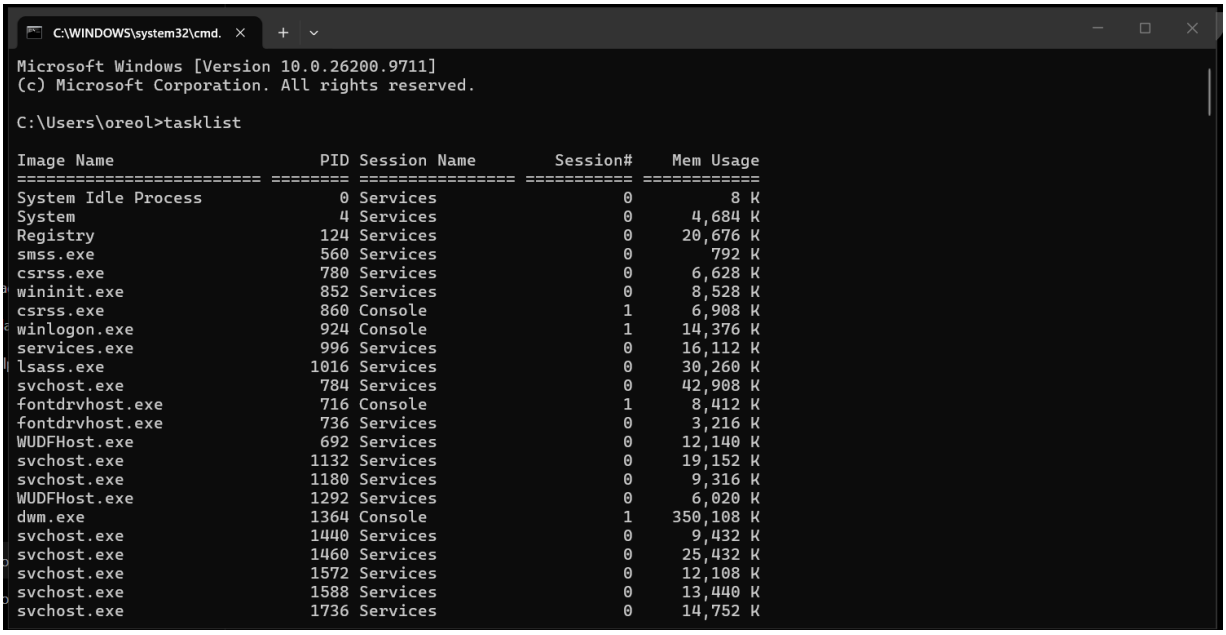


Figure 7. Command Prompt tasklist output showing running process names, PIDs and memory usage.

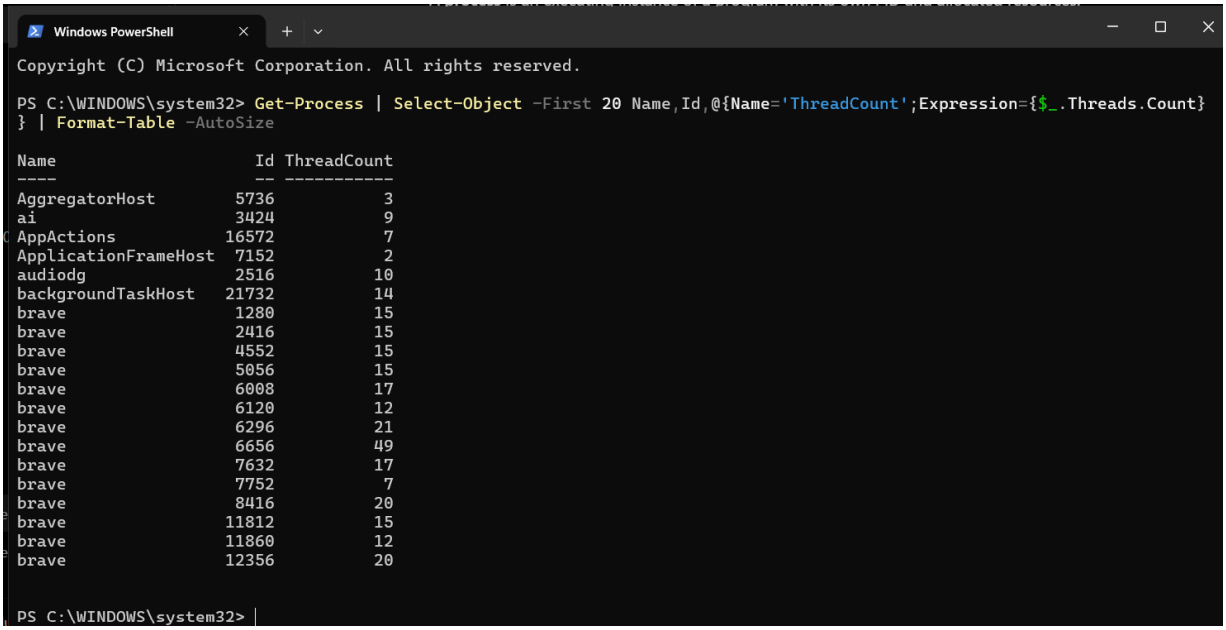


Figure 8. PowerShell Get-Process output showing process IDs and thread counts.

B2. Explain the difference

A program is software stored on a computer. A process is a running program with its own PID. A thread is a unit of work within a process. A process can have several threads.

The Brave browser has several running processes in Figure 8. Each process has its own PID and thread count.

Process names, PIDs and thread counts help investigators identify running software and check for unusual activity during malware investigations.

The screenshots identify running processes, their PIDs and thread counts.

PART C — DISK SECTOR AND DISK SIZE CALCULATION

C1. Reproduce the slide calculation

Given: 400 sectors per track, 12 heads, 17,000 cylinders and 512 bytes per sector.

Item	Calculation	Result
Total sectors	400 × 12 × 17,000	81,600,000 sectors
Total bytes	81,600,000 × 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 ÷ 1,024	40,800,000 KB
Megabytes	40,800,000 ÷ 1,024	39,843.75 MB
Gigabytes	39,843.75 ÷ 1,024	38.91 GiB (rounded)

C2. Calculate the sizes of partitions 1 and 2

Disk 0 has 512 bytes per logical sector. The partition details below come from the read-only PowerShell output.

Partition / type	Start sector	Sector count	Bytes/sector	File system
1 — System	2,048	409,600	512	Not shown
2 — Reserved	411,648	32,768	512	Not applicable (MSR)

Formula: Partition size = sector count × bytes per sector. Size in GiB = bytes ÷ 1,024 ÷ 1,024 ÷ 1,024.

Partition 1: 409,600 × 512 = 209,715,200 bytes.
209,715,200 ÷ 1,024 ÷ 1,024 ÷ 1,024 = 0.1953125 GiB (200 MiB).

Partition 2: 32,768 × 512 = 16,777,216 bytes.
16,777,216 ÷ 1,024 ÷ 1,024 ÷ 1,024 = 0.015625 GiB (16 MiB).

Sector check: Partition 1 starts at 2,048 and ends at 411,647. Partition 2 starts at 411,648 and ends at 444,415. These values match the PowerShell output.

```
PS C:\WINDOWS\system32> $disk = Get-Disk -Number 0
PS C:\WINDOWS\system32> Get-Partition -DiskNumber 0 |
>> Where-Object { $_.PartitionNumber -in 1,2 } |
>> ForEach-Object {
>>     $start = [uint64]($_.Offset / $disk.LogicalSectorSize)
>>     $count = [uint64]($_.Size / $disk.LogicalSectorSize)
>>     [pscustomobject]@{
>>         Partition = $_.PartitionNumber
>>         StartSector = $start
>>         EndSector = $start + $count - 1
>>         SectorCount = $count
>>         BytesPerSector = $disk.LogicalSectorSize
>>         SizeBytes = $_.Size
>>         SizeGiB = [math]::Round($_.Size / 1GB, 8)
>>     }
>> } | Format-Table -AutoSize

Partition StartSector EndSector SectorCount BytesPerSector SizeBytes   SizeGiB
--------- ----------- --------- ----------- -------------- ---------   -------
        1        2048    411647      409600            512 209715200 0.1953125
        2      411648    444415       32768            512  16777216  0.015625


PS C:\WINDOWS\system32>
```

Figure 9. Read-only PowerShell output showing the start sector, end sector, sector count and calculated sizes for partitions 1 and 2.

PART D — SAFE PARTITION TABLE INSPECTION

D1. Windows method

Disk Management shows Disk 0 as an online 238.46 GB disk, with an EFI System partition, C: partition and Recovery partition. The Microsoft Reserved partition is shown in the PowerShell and DiskPart listings.

Disk	Partition	Type	Size (DiskPart)
0	1	System	200 MB
0	2	Reserved	16 MB
0	3	Primary	237 GB
0	4	Recovery	918 MB

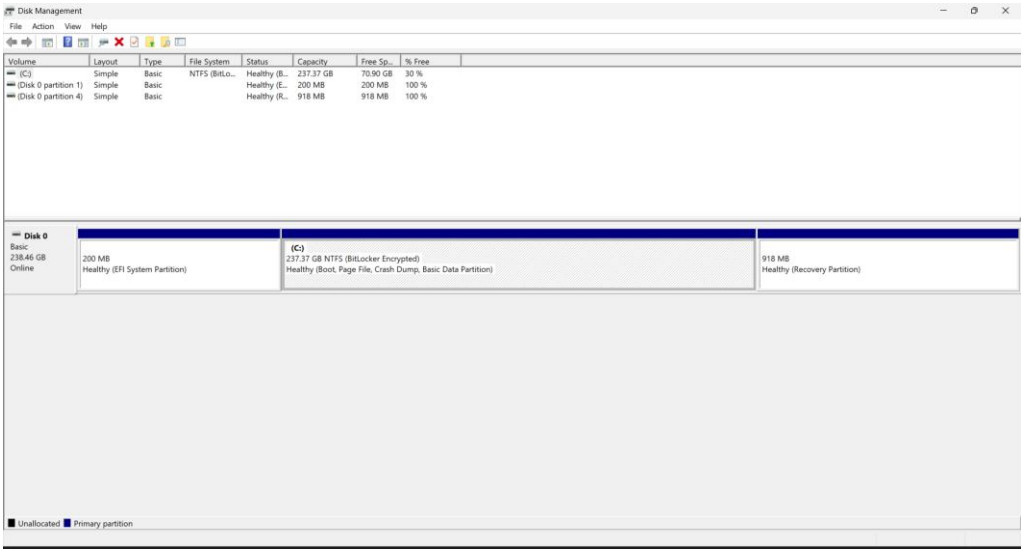


Figure 10. Disk Management showing Disk 0 and its visible partitions.



Figure 11. DiskPart read-only list disk and list partition output.

PART E — MBR, GPT AND BOOT PROCESS EXPLANATION

E1. Explain the boot process

1. Power-on and POST: When the PC is switched on, it receives power and the firmware runs the Power-On Self-Test (POST). This checks basic hardware, including the memory and processor, before booting continues.

2. BIOS/UEFI and firmware settings: The firmware reads saved settings, such as the boot order, and selects a device to start from. Older BIOS systems use CMOS-backed settings; this PC uses UEFI firmware.

3. MBR/GPT and EFI boot records: On older BIOS systems, boot code in the MBR starts the boot process. On this UEFI/GPT system, a firmware boot entry points to Windows Boot Manager on the EFI System Partition.

4. Boot manager and loader: Windows Boot Manager selects the Windows installation and starts the Windows OS loader. The loader prepares the files needed to start Windows.

5. Kernel loading: The loader brings the Windows kernel and essential drivers into memory. The kernel then takes control of the system's main operations.

6. Operating system control: Windows starts system services and the sign-in process. Once signed in, the user can run programs and access files.

Forensic relevance: Firmware settings and boot entries can show how the PC is set to start. Boot files and Windows logs may provide evidence about startup activity.

My system uses UEFI, not Legacy BIOS. This is shown in the System Information screenshot in Part A. The disk uses GPT, as shown by the Get-Disk output in Part A.

E2. Compare MBR and GPT

Feature	MBR	GPT
Age / design	Older partitioning scheme.	Newer partitioning scheme commonly used with UEFI.
Size limitation	Typically supports disks up to 2 TiB with 512-byte sectors.	Supports much larger disks.
Partition information	Partition entries are stored in the first sector alongside boot code.	Uses a primary and backup GPT header and partition-entry array.
Forensic issue	Damage to sector 0 can affect partition interpretation.	Backup GPT structures may help check or recover partition information.

PART F — FILE SYSTEM IDENTIFICATION AND INTERPRETATION

The Get-Volume results in Part A show two NTFS volumes:

Volume / partition	File system	Size
C: (Partition 3)	NTFS	237.37 GB

Recovery (Partition 4)	NTFS	918 MB

Screenshot evidence: The Get-Volume screenshot in Part A shows the file systems. The Get-Partition screenshot identifies the partition numbers.

A file system organises files and stores details such as file names, locations and timestamps. This matters in digital forensics because investigators can use these details to find files and examine when activity happened. NTFS records file information in the Master File Table (MFT).

PART G — FINAL PRACTICAL ASSIGNMENT

Partition table evidence: Disk 0 uses GPT. The Get-Partition output in Part A, the sector calculation screenshot in Part C (Figure 9), and the DiskPart partition listing in Part D (Figure 11) provide the values used below.

Partition 1 — EFI System Partition

1. Identify: Disk 0, partition 1, System (EFI), shown as 200 MB in DiskPart and Disk Management (Figures 10–11).
2. Verify sectors: Start sector = 2,048; end sector = 411,647 (Figure 9).
3. Sector count = 411,647 − 2,048 + 1 = 409,600 sectors.
4. Size = 409,600 × 512 bytes = 209,715,200 bytes.
5. Size in GiB = 209,715,200 ÷ 1,073,741,824 = 0.1953125 GiB.

Partition 2 — Microsoft Reserved Partition

1. Identify: Disk 0, partition 2, Reserved (MSR), shown as 16 MB in DiskPart (Figure 11).
2. Verify sectors: Start sector = 411,648; end sector = 444,415 (Figure 9).
3. Sector count = 444,415 − 411,648 + 1 = 32,768 sectors.
4. Size = 32,768 × 512 bytes = 16,777,216 bytes.
5. Size in GiB = 16,777,216 ÷ 1,073,741,824 = 0.015625 GiB.

Comparison with Windows-reported sizes

Partition	Calculated size	Windows-reported size
1 — EFI System	0.1953125 GiB (200 MiB)	200 MB
2 — MSR	0.015625 GiB (16 MiB)	16 MB

The displayed sizes agree after unit conversion. Windows labels these sizes as MB, while the sector calculations give 200 MiB and 16 MiB. The smaller GiB values are due to converting to a larger unit, not missing partition space.

CONCLUSION

In this lab, I identified my PC specifications, checked running processes and inspected disk partitions using Windows tools. I calculated the sizes of partitions 1 and 2 from their sector counts and confirmed that the system uses UEFI, GPT and NTFS volumes. I learned that system details, process records, partition information and file system metadata can help investigators understand computer activity.